INCH - POUND

MIL-I-46852C
27 May 1991
SUPERSEDING
MIL-I-46852B
8 MAY 1973

MILITARY SPECIFICATION

INSULATION TAPE, ELECTRICAL, SELF ADHERING UNSUPPORTED SILICONE RUBBER

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for an unsupported silicone rubber tape which will adhere to itself at room temperature. The tape is suitable for use at temperature from -90° to 260° Celsius (C).

1.2 Classification. The tape shall be furnished in the following types as specified.

Type I - Rectangular cross-section

Type II - Triangular cross-section with colored guide lines

2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970

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2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

FEDERAL

PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-T-22085	-	Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing
MIL-I-24391	-	Insulation Tape, Electrical, Plastic Pressure-Sensitive
MIL-T-43036	-	Tapes, Pressure-Sensitive, Adhesive, Plastic Film (for Sealing Fiber Containers and Cans)

STANDARDS

FEDERAL

FED-STD-601	-	Rubber: Sampling and Testing, Federal Test Method for

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified,

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the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM D 149	-	Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies, Standard Test Method for
ASTM D 1458	-	Fully Cured Silicone Rubber-Coated Glass Fiber and Tapes for Electric Insulation, Standard Test Methods for
ASTM D 3759	-	Tensile Strength and Elongation of Pressure-Sensitive Tapes, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The tape shall be made of a self-adjusting silicone rubber material which will cure into a homogeneous mass. When wrapped or plied on itself, it will cure to a homogeneous mass at room temperature (21° to 27°C) in 48 hours or in 4 hours at 177°C.

3.3 Performance and product characteristics. The unsupported silicone rubber tape shall have the following performance and product characteristics:

3.3.1 Dimensions and color. The tape shall be furnished in the following thicknesses, widths, and colors as specified in table I and table II.

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TABLE I. Type I dimensions.

Thickness	0.010, 0.020, 0.030 ± 0.002 inches
Width	0.50, 0.75, 1.0, 1.25 and 1.50 inches

TABLE II. Type II dimensions and color.

Thickness (Inches)	Width (Inches)	Color
0.020 ± 0.002	1.00	Blue
0.030 ± 0.002	1.00	White
0.040 ± 0.002	1.00	Green
0.050 ± 0.002	1.25	Yellow
0.060 ± 0.002	1.50	Blue
0.080 ± 0.002	1.50	White
Thickness edge, not less than 0.007 inches and not greater than 0.009 inches		

3.3.1.1 Rolls. Unless otherwise specified in the contract or purchase order, the tape shall be supplied in rolls of specified thickness and width (see 3.3.1), wrapped on a core with tape interleaved with quick-release film. The release film shall be of sufficient width to preclude tape edges from fusing during packaging, packing, and storage. Unless otherwise specified in the contract or purchase order, the tape shall be supplied in rolls not less than 12 yards in length.

3.4 Physical properties. The physical properties of the material shall be as specified in table III.

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TABLE III. Physical properties.

Property		Value	Method paragraph
Tensile strength Pound force per square inch (lbf/in^2)		700 minimum	4.6.1
Elongation Percent		300 minimum	4.6.2
Water Absorption Percent by Weight		3 maximum	4.6.3
Tape Thickness			
Dielectric Strength	0.010 inch	400 minimum	4.6.4
Volts/milimeter	0.020 inch	400 minimum	
	0.030 inch	400 minimum	
	0.040 inch	300 minimum	
	0.050 inch	275 minimum	
	0.060 inch	250 minimum	
	0.080 inch	200 minimum	
Volume resistivity, ohms/circular milimeter/foot		1×10^{13}	4.6.5
Bond strength, 1 inch width		2 lbs minimum	4.6.6

3.5 Storage life. Tape stored in unopened containers at a temperature of not greater than 37.7°C shall be capable of meeting all the requirements of this specification for a period of 12 months from date of manufacture. For storage life extensions see paragraph 6.3.

3.6 Workmanship. The tape shall be uniform in quality and shall be free from splits, cracks, foreign material, and other defects that would affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.7.5 Solvent resistance. The solvent resistance tests for

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified, (see 6.2), a sample shall be subjected to first article inspection in accordance with tests specified herein.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in table IV.

TABLE IV. Quality conformance inspection.

Inspection	Requirements	Test paragraph
Tensile strength	Table III	4.6.1
Elongation percent	Table III	4.6.2
Water absorption percent by weight	Table III	4.6.3
Dielectric strength	Table III	4.6.4
Volume resistivity	Table III	4.6.5
Bond strength	Table III	4.6.6

4.5 Inspection provisions.

4.5.1 Lot size. An inspection lot, as far as practicable, shall consist of all the rolls of tape of the same thickness, width, and color, produced under essentially the same conditions, and offered for inspection at one time.

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4.5.2 Sampling.

4.5.2.1 For examination. A random sample shall be selected from each lot in accordance with table V.

4.5.2.2 For acceptance testing. A random sample shall be selected from each lot in accordance with table V.

TABLE V. Sampling plan for examination and acceptance testing.

Lot Size	Sample Size
2 to 8	All
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.5.3 Examination.

4.5.3.1 Visual examination. Examination of the samples specified in 4.5.2.1 shall be conducted for the purpose of determining compliance with the requirements specified in 3.3 and 3.5.

4.5.4 Acceptance testing. Testing of the samples specified in 4.5.2.2 to determine compliance with the following characteristics shall be conducted in accordance with the test paragraphs of table VI.

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TABLE VI. Acceptance testing

Inspection	Test paragraph
Tensile strength	4.6.1
Elongation	4.6.2
Water absorption	4.6.3
Dielectric strength	4.6.4
Volume resistivity	4.6.5
Bond strength	4.6.6

4.6 Test methods and procedures. The tape shall be subjected to the following or equivalent tests:

4.6.1 Tensile strength. The tensile strength shall be measured in accordance with ASTM D 3759.

4.6.2 Elongation. Elongation shall be measured in accordance with ASTM D 3759.

4.6.3 Water absorption. Water absorption shall be measured in accordance with FED-STD-601, Method 6251, using the tape as received.

4.6.3.1 Preparation of test specimen. For type II (triangular cross section) tape, three strips of tape shall be lapped together as shown in figure 1 to provide material for a dumbbell shaped diecut test specimen with uniform thickness.



FIGURE 1. Preparation of tape specimen for water absorption test.

4.6.4 Dielectric strength. Dielectric strength shall be measured in accordance with ASTM D 149.

4.6.5 Volume resistivity. The volume resistivity shall be measured in accordance with ASTM D 1458.

4.6.6 Bond strength. Two sample strips 1 by 12 inches shall be placed face to face with the lining removed. This laminate shall be placed between two layers of glass cloth and rolled twice with a 4.5 pound roller at a rate of 12 inches per minute. The bond strength of the tape laminate shall be not

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less than 2 pounds when pulled apart with a tensile tester at a crosshead rate of 10 inches per minute.

4.6.7 Material age. Determine the age of the material from date of manufacture (see 3.5 and 5.3).

4.6.8 Inspection of preparation for delivery. An examination of the preparation for delivery shall be performed to determine compliance with the requirements of Section 5 of this specification. The lot shall consist of items, packages or shipping containers, as applicable. The sample unit shall be one item, one package or one shipping container. The sample size and accept number shall be in accordance with table VII.

TABLE VII. Sampling plan for inspection of preparation for delivery.

Lot Size	Sample Size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 to 35,000	29
35,001 to 150,000	29
150,001 to 500,000	29
500,001 and over	29
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.6.9 Storage life. The supplier shall certify that the material will meet all requirements of this specification after being subjected to the storage life conditions as specified in 3.5.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Unit packaging. Each roll of tape shall be packaged in a snug fitting container conforming to PPP-C-96, Type IV, Class 3, with a slip cover closure. Sealing shall be accomplished by encircling the closure seam with a one inch width of tape conforming to MIL-T-22085, Type 2, MIL-T-43036 or MIL-I-24391. The tape shall overlap not less than 0.5 inch. When more than a single roll of tape is specified for the unit package quantity, a non-adhering separator shall be placed between the rolls.

5.2 Packing. Packing shall be Level A or B as specified (see 6.2).

5.2.1 Level A. A specified quantity (see 6.2) of tape packaged as described in 5.2 shall be inserted in a snug fitting container conforming to PPP-B-636, class WR. The greatest gross weight and dimensional limitations for special requirement containers shall apply. When the specified quantity exceeds a net weight of 60 pounds, the shipping container shall conform to PPP-B-640, class 2, grade A. Maximum packed gross weight shall be not greater than 200 pounds closure and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Level B packing shall be the same as Level A.

5.3 Marking. In addition to any marking specified in the contract or order (see 6.2), unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and in addition shall include the following:

a. This specification number, revision level, and type

b. Manufacturer's name and material designator

c. Quantity and dimensions

d. Date of manufacture and expiration date

e. Storage requirement statement: "Store below 37.7°C.".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tape is intended to be used as electrical insulation and as a thermal protection barrier for components subjected to missile blast environments.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Type required (see 1.2)

d. Type II, color required (see 3.2.1)

e. Width, length, and thickness of tape (see 5.3)

f. Level of packaging and level of packing required (see 5.1 and 5.2)

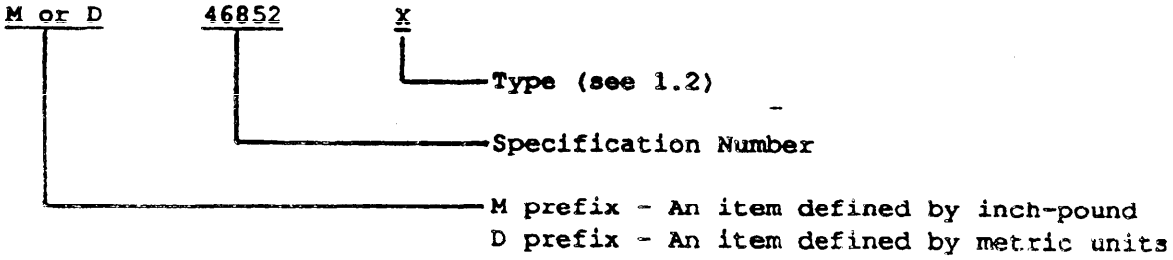
g. Whether special marking is required (see 5.3)

h. Whether a first article is required (see 3.1).

6.3 Storage life extension. Tape which has exceeded storage life may be examined for compliance with paragraph 3.4, table III (Physical Properties), in accordance with paragraph 4.4, and table IV (Quality Conformance Inspections). Acceptable inspection test results shall constitute storage life extension for an additional 6 months.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Part or identification number.



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6.6 Subject term (keyword) listing.

Dielectric breakdown
Dielectric strength
Pressure sensitive materials
Tensile strength

6.7 Metrication. See appendix.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MI
Air Force - 85

Preparing Activity:
Army - MI

(Project No. 5970-1067)

Review Activity:
Air Force - 99
DLA - GS
DNA - DS

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APPENDIX

INSULATING TAPE, ELECTRICAL, SELF ADHERING
UNSUPPORTED SILICONE RUBBER, ACCEPTABLE METRIC SIZES

10. SCOPE

10.1 Scope. This appendix establishes the requirements for acceptable hard metric (SI) sizes of electrical insulating tapes which shall conform to all requirements and quality assurance provisions (Sections 3 and 4 of this specification), except as noted in this appendix. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

10.2 Classification. Tapes conforming to standard metric dimensions shall be furnished in the following types as specified.

Type I-M - Rectangular Cross Section

Type II-M - Triangular cross-section with colored guidelines

20. APPLICABLE DOCUMENTS

20.1 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 380 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

30. REQUIREMENTS

30.1 Dimensions and color. When specified (see 6.2.e) the tape shall be furnished in the following thicknesses, widths, and colors as specified in table I-A and table II-A.

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APPENDIX

TABLE I-A. Type I-M dimensions.

Thicknesses	0.250, 0.500, 0.750, ± 0.050 millimeters (mm)
Widths (mm)	15.0, 20.0, 25.0, 30.0, 40.0 ± 0.5 mm

TABLE II-A. Type II-M dimensions and color.

Thickness (mm)	Width (mm)	Color
0.500 ± 0.050	25.0 ± 0.5	Blue
0.750 ± 0.050	25.0 ± 0.5	White
1.000 ± 0.050	25.0 ± 0.5	Green
1.300 ± 0.050	32.0 ± 0.5	Yellow
1.500 ± 0.050	38.0 ± 0.5	Blue
2.000 ± 0.050	38.0 ± 0.5	White
Thickness edge: not less than 0.200 mm; not greater than 0.230 mm		

30.2 Physical properties. The physical properties of the tape shall be as specified in 3.4 and Table III, except dielectric strength shall be in accordance with Table III-A, when measured in accordance with ASTM D 149 (see 4.6.4); and bond strength shall be determined in accordance with 40.1.

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APPENDIX

TABLE III-A. Dielectric strength for metric product.

Property	Value
Dielectric strength: Tape Thickness (mm)	Volts/mil 1/
0.250	400 min
0.500	400 min
0.750	400 min
1.000	300 min
1.300	275 min
1.500	250 min
2.000	200 min
Bond strength: Tape width (mm)	kgf (min)
25	.9 kgf

1/ This requirement is expressed the same as in Table III, (volts per thousandths inch thickness).

30.3 Lengths. Tape length for both types shall be 10 meters, +1 m, -0 m.

40. QUALITY ASSURANCE PROVISIONS

40.1 Bond strength. Two sample strips 25 mm by 305 mm shall be placed face to face with the lining removed. This laminate shall be placed between two layers of glass cloth and rolled twice with a 2 kg roller at a rate of 305 mm per minute. The bond strength of the tape laminate shall be not less than .9 kgf when pulled apart with a tensile tester at a crosshead rate of 250 mm per minute (4.16mm per second).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-46852C	2. DOCUMENT DATE (YYMMDD) 910527

3. DOCUMENT TITLE
INSULATION TAPE, ELECTRICAL, SELF ADHERING UNSUPPORTED SILICONE RUBBER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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